Individual Demonstration Instructions and Rubric

Preparing for the demonstration

Before this demonstration lab session, you are expected to have updated your own code as follows:

- Although code will not be directly graded, you must be able to show the code you are running, re-upload it if requested by the lecturers, and answer any questions we ask you about it.
- Your code should consist of
    - Your low-level STM32 HAL code (as demonstrated in your Individual Demonstration in Semester 1, and which you updated for Prototype Design 3)
    - The high-level code that you submitted in Prototype Design 5b (e.g. DSP, state machines etc) – but without the testing harnesses.
    - This could should all be wrapped within an appropriate set of FreeRTOS tasks
- The code should be your individual contribution, and should not include any code of other team members.
    - If you think you have grounds for an exemption (e.g. you need to be able to detect a button press to run your tests and your team mate wrote that code), then confirm this with lecturers in advance.
- If your high-level code is not easy to demonstrate without full integration of the system, then discuss this with lecturers
    - We will try to come up with a feasible way to demonstrate those pieces of code, so that you can receive grades for that functionality during the demonstration.
- A higher level of integration is expected than in the semester 1 demonstrations
    - Your own pieces of code should be working together, but are not expected to be working with those of your team mates.
    - As you are using an RTOS, where appropriate you should show that multiple parts of your system can operate in parallel.
- You must have a single firmware program that exercises all functions
    - There should be no re-compiling and re-uploading during the demonstration, all tests should be done with one with the same set of firmware.
    - If your subsystem needs to transfer data to/from a PC or mobile phone (e.g. to test ethernet, Bluetooth, USB etc), then you are free to use existing or custom programs/scripts

During the demonstration

Each student will have 20 minutes to demonstrate their subsystem(s) to a lecturer. You must be ready to start demonstrating at your allocated time, schedule will be on Moodle. Your task is to

- Demonstrate that you have successfully controlled all of the relevant functions of your hardware
    - This should be similar to what you showed during your semester 1 individual demonstration – although quantitative measurements are not required.
- You should demonstrate integrated functionality as much as possible, for example:
    - If you have buttons, a menu system, and a display, you should show them all working together

    - If you have an SD card reader and an MP3 decoder, then find a way to demonstrate that the audio files were correctly read from the SD card and decoded into raw audio samples.
- You are responsible for ensuring that the procedure you use convincingly demonstrates that your subsystem(s) are working as expected.
    - Lecturers reserve the right to modify your tests to ensure that results are genuine.
    - Your demonstration should not rely on inspecting the code – correct operation should be made clear from the observed behavior of the system.
    - To demonstrate that your own tasks are well-behaved multitasking, you can instrument them using STM32 debugger/profiler, or add some other tasks doing dummy work in the background (e.g. continuously printing to the serial monitor)
- You are free to use any appropriate tools or instruments to test functionality of your board and code, including:
    - Oscilloscopes, signal generators, power supplies, differential probe, electronic load
    - Sound level meter and sound calibrator
    - Load resistors and inductors
    - Laptops (including connections over the serial monitor), mobile phones, other dev boards, microcontrollers, data acquisition system
    - The STM32 debugging and profiling tools
- During the demonstration, you will be asked some questions about your subsystem(s)
    - You should be able to answer basic questions about your subsystem(s) or your measurement procedure from your head.
    - For more complex questions, it is acceptable to refer to your design documents, systems engineering document, datasheets, application notes, or your code.
    - It is not permitted to use any search engine or AI tool when answering questions – all relevant documents should be readily accessible and findable on your PC.
    - If your subsystem(s) are not working as expected, you will need to explain why not. This explanation should be backed up by some evidence (e.g. measurements or referring to your schematic)

ZEIT4230 Individual Demonstration Grading Sheet

Student Name: Grader Name:

Overall Grade: /100 Date:

Criterion	Grade (or Y/N)	Comment
All parts on the circuit board needed for the student's design to work are functioning (Y/N) • The STM32 microcontroller and associated circuitry function correctly • All hardware that the student was responsible for in the design function correctly • A reasonable attempt to assemble all hardware on the board (including parts others are responsible for) • (A grade penalty may be applied if above criteria are not met)	Y/N	
Power rails needed to power the student's own circuitry (including microcontroller) are powered correctly when using • The 24VAC power supply for a mains-powered system, or • Batteries for a battery-operated system • (A grade penalty may be applied if above criteria are not met)	Y/N	
Level of functionality demonstrated in higher level application code • Convincing demonstration that your own DSP, state machines, menu systems or other high level program logic operate correctly • Demonstrate the ability to perform multiple operations in parallel, or to provide timely operation even when dummy background tasks are running • Evidence should be based on demonstrated functionality of the system or by instrumenting registers and variables, but not by inspecting the code	(60% weight)	
Student understanding when answering questions • Ability to explain their own hardware and software in detail • High level understanding of other group member's hardware • Including ability to find information in relevant sources. • (If applicable) Student's understanding of why any part of the board is not functioning (potentially including parts that belong to other group members), backed by evidence.	(40% weight)	